



Smart Contract Audit

Security Assessment

Audit Dashboard Link

<https://app.chinaudits.io/project/octoswap-1?cid=110>

OctoSwap

V1 Periphery



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Project Details

CODEBASE
Upload

NETWORK
N/A

LANGUAGE
Solidity

NAME
OctoSwap

TOTAL SUPPLY
N/A

OWNER
N/A

WEBSITE
<https://octo.exchange/>

UNIT TESTS
N/A

FORK
N/A

ABOUT THE PROJECT

OctoSwap is a decentralized exchange on Monad that combines traditional automated market maker pools with concentrated liquidity pools in a single trading system. Its smart routing engine sources liquidity across both models to provide efficient execution for users. The platform includes swap functionality, liquidity management, and ecosystem features such as gasless approvals, MEV protections, incentives, and a points system.

Protocol Inheritance Analysis:

OctoSwap's contracts are verified to be directly forked from Uniswap contracts. All functional components were reviewed to confirm that the implemented logic matches Uniswap's original design. The differences identified are cosmetic in nature and do not alter the core functionality or expected behavior of the protocol.

Social Media Links

Linktree	N/A
Telegram	N/A
Facebook	N/A
Medium	N/A
X	• https://x.com/OctoSwapDex
Instagram	N/A
Youtube	N/A
Github	N/A
Reddit	N/A
Discord	N/A
Tiktok	N/A
LinkedIn	N/A
Flooz	N/A
Coingecko	N/A
Coinmarketcap	N/A

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Vulnerability Summary

● Critical

Critical risks are those that affect the platform's safe operation and must be resolved before launch. Users should avoid investing in any project with unresolved critical risks.

● High

High risks include centralization issues and logical errors. These risks can potentially result in the loss of funds or control over the project under certain conditions.

● Medium

Medium risks might not directly threaten users' funds, but they can impact the platform's overall functionality.

● Low

Low risks are similar to the above categories but on a smaller scale. They typically do not compromise the project's overall integrity but may lead to less efficient solutions.

● Informational

Informational errors are recommendations aimed at improving code style or aligning operations with industry best practices. These usually do not affect the code's overall functionality.

● Optimization

Optimization issues refer to inefficiencies in the code or smart contract logic that could be improved for better performance, reduced gas costs, or enhanced scalability. These issues do not impact security but can make the system more cost-effective and efficient.

Note - The provided audit report thoroughly examines the security aspects of the smart contract employed in the project, encompassing potential malicious manipulation of the contract's functions from external sources. However, it's important to note that this analysis does not incorporate functional or unit testing of the contract's logic. Therefore, we cannot ensure the absolute correctness of the contract's logic, including internal calculations within the formulae utilised in the contract.

Overview

Score



The score is solely based on audited code findings and is not to be used as financial advice.

In-Scope Files

The team provided us with files to review during security audits. This audit covered the following files listed below with their respective SHA-1 Hash.

File Name	SHA-1 Hash
DeflatingERC20.sol	97bdb52f81eebe9db85eb7e32d6554f54f173ccf
ERC20.sol	8dc751de5bd48e54734f016976d7ae66f61e9bed
IERC20.sol	b7d011aaabd34898ee60760996cb702e7b2ca855
IUniswapV2Router01.sol	f150379f2a39f6d992f9a0aa35915ea1f64658d9
IUniswapV2Router02.sol	0d2ae53fa0c621cbc7e0e46e506c562068037f6c
IWETH.sol	6a25dd53c8494e3aef3a520f17e00608b529f061
OctoswapRouter.sol	964f8c763e973b6690a07abc77aaddbc5f5362b4
RouterEventEmitter.sol	f65c6faa7061980ab64d138abdd945c7a3f99ecf
SafeMath.sol	123c932c8701c1178d049c82339bc68cd3c61d18
UniswapV2Library.sol	275524d3f119415856ee7de7f37969e73f9f923c
UniswapV2LiquidityMathLibrary.sol	0ace278f6d7278e48799c40653e87b8629ae7555
UniswapV2OracleLibrary.sol	e6d753b6992df2cf7a11e57eddb350c2026d4e8b
WETH9.sol	0e2919ba8f8a85a49d28e505892e2834e0591749

Please note that files with hash values different from those listed in this table have been changed after security checks, intentionally or unintentionally, as a particular hash value may indicate a changed state or potential vulnerabilities not checked in this scan.

Note for Investors: We only examined agreements indicated in the indicated ratings. No contracts associated with the project beyond this range have been audited, therefore, we cannot provide insight or assume responsibility for their security



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External/Public functions

External/public functions can be invoked outside of the contract, i.e., accessed by other contracts or external accounts on the blockchain. These functions are identified using external or public visibility modifiers in the function declaration.

State variables

State variables are stored on the blockchain as part of the contract conditions. They are declared at the contract level and can be accessed and changed by any action in the contract (except with modifiers like `onlyOwner`, etc.). State transitions can be described using a visibility modifier, such as `public`, `private`, or `internal`, which refers to the access to the transition.

Components

Contracts	5
Libraries	4
Interfaces	4
Abstract	0

Exposed Functions

Public	83
Payable	14
External	66
internal	86
Private	4
Pure	18
View	19

State Variables

Total	25
Public	25

Capabilities

Solidity Versions observed	=0.6.6, >=0.5.0, >=0.6.2
Transfers ETH	`yes`
Can Receive Funds	yes
Uses Assembly	yes (2 asm blocks)
Has Destroyable Contracts	No

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Audit Information Strategies

While our projects undergo rigorous testing to address all known vulnerabilities, it's important to provide further clarification and understanding by outlining specific vulnerabilities that have been identified. This additional information enhances transparency and helps stakeholders comprehend the security measures taken and any potential risks mitigated.

Title	issue
Reentrancy	Improper Enforcement of Behavioral workflow
Unexpected Ether Balance	Improper Locking
Code with no effects	Irrelevant Code
Flash Loan Attacks	External Oracle Manipulation
Sandwich Attacks	A Form of Front Running
Write to arbitrary storage location	Improper Write to Arbitrary Storage Location
Variable Shadowing	Improper Coding Standards
Unprotected SELF-DESTRUCT	Improper Access Control
Potential Honeypot	Improper Function
Unprotected ETH withdrawal	Improper Access Control
Outdated Compiler Version	Using components with Known vulnerabilities
Weak Source of Randomness from Chain Attributes	Use of Insufficiently Random values
Unsafe use of libraries	Improper Implementation
Wrong Implementation of Token Standards	Improper Coding Standards

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Auditing Strategy and Techniques Applied

All Projects at ChainAudits undergo our in-house developed “4-Eye” method. This process ensures that the code is analyzed not by a single auditor but by our entire technical team. Moreover, this is accomplished most efficiently, leaving no stone unturned.

We manually check every file, line by line. Automated tools are used solely to help us achieve faster and better results.

Code Analysis Methodology

The auditing process follows a routine series of steps:

- Manual Code Review:
 - Evaluate the overall structure and organization of the smart contract code line-by-line
 - Review the implementation of access control mechanisms to ensure that sensitive functions are appropriately restricted to authorized users.
 - Ensure that critical vulnerabilities are adequately tested and that edge cases and boundary conditions are covered.
- Static Code Analysis:
 - Utilise static analysis tools to scan the codebase for vulnerabilities and security issue
 - Identify common vulnerabilities like reentrancy, integer overflow/underflow, and unchecked external calls.
 - Evaluate compliance with coding standards and best practices, ensuring adherence to security guidelines.
- Code Structure and Architecture Review:
 - Analyze the overall structure and architecture of the smart contract code.
 - Assess the modularity, readability, and maintainability of the code.
 - Review the separation of concerns and adherence to design patterns for robustness and security.
- Security Best Practices Evaluation:
 - Evaluate the implementation of security best practices, including access control mechanisms and input validation.
 - Verify proper error handling to mitigate potential vulnerabilities and ensure contract robustness.
 - Check for gas optimization techniques to enhance efficiency and reduce transaction costs.
- External Dependency Assessment:
 - Assess the integration with external contracts, libraries, or services.
 - Review the security of external dependencies and their impact on the overall security of the smart contract codebase.
 - Ensure secure interactions with external components to prevent vulnerabilities and attack vectors.
- Post-Analysis Support:
 - Offer support and guidance to assist the development team in addressing identified vulnerabilities.
 - Collaborate with stakeholders to ensure effective implementation of recommended security enhancements.
 - Provide ongoing assistance and consultation to maintain the security of the smart contract codebase.

General Findings - Type: ERC-20

Can the owner mint tokens arbitrarily?

The owner may have the ability to mint new tokens at any time, potentially diluting the token supply.

No

Can the owner burn tokens from any address without approval?

If the owner can burn tokens from user addresses without the holder's approval, it could lead to unjust token loss.

No

Can the owner pause or freeze token transfers?

The owner may have the authority to pause all transfers, which could prevent users from accessing their funds when needed.

No

Can the owner blacklist or whitelist addresses?

The owner may be able to control which addresses can hold or transfer tokens, potentially leading to censorship or arbitrary exclusion of participants.

No

Are there any integer overflow or underflow vulnerabilities?

The contract should ensure that all arithmetic operations are safe from overflow or underflow issues, especially if using older Solidity versions where these checks are not automatic.

No

Is there improper handling of Approvals?

The approve function should properly mitigate risks of double-spending when allowance is modified and then used in separate transactions.

No

Disclaimer

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ChainAudits reports aim to identify successful audit processes to help our clients improve the quality of the code and manage the risks associated with cryptographic tokens and blockchain technology. It is important to understand that blockchain technology and cryptographic assets pose significant ongoing risks. Each company and individual should conduct its own due diligence to maintain consistent safety measures. ChainAudits makes no representations about the security or performance of the technologies we audit.

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